

Phillip Lippe

PhD student focusing on (causal) representation learning and generative pretraining with experience in large-scale training. Looking for Research Scientist positions based in Europe and US with starting dates from October 2024 on.

Nationality German
🔗 [phillippe.github.io](https://github.com/phillippe)

Languages German (native), English (C1)
in [linkedin.com/in/phillip-lippe](https://www.linkedin.com/in/phillip-lippe)

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EXPERIENCE

Student Researcher

Google DeepMind

📅 Aug 2023 – Nov 2023 📍 Amsterdam, Netherlands

- Research in generative multimodal pretraining with positive transfer across modalities; hosted by [Mostafa Dehghani](#)
- Parallelized training of models with up to 4 billion parameters
- Results internally used in the [Gemini](#) foundation model project

Research Intern

Microsoft Research

📅 Mar 2023 – May 2023 📍 Amsterdam, Netherlands

- Research in neural PDE solvers for scientific simulations like fluid dynamics; hosted by [Johannes Brandstetter](#)
- Results published in *PDE-Refiner: Achieving Accurate Long Roll-outs with Neural PDE Solvers* at NeurIPS 2023 ([pdf](#))

Student Research Intern

Daimler AG, Mercedes-Benz

📅 Dec 2016 – Mar 2017
Apr 2018 – Sep 2018 📍 Stuttgart, Germany

- Research in deep learning for real-time multi-class object detection in complex urban traffic scenes (autonomous driving)
- Bachelor Thesis: *Hierarchical Multi-label Object Detection of Rare Classes for Autonomous Driving* ([pdf](#))

Student Research Intern

Mercedes-Benz Research and Development NA

📅 May 2017 – Aug 2017 📍 Sunnyvale, United States

- Research in deep generative models for predicting agent behavior in traffic scenarios for autonomous driving ([report](#))
- Performed agile software development with Scrum

PROJECTS

Google Developer Expert, Machine Learning

- Member of the ML GDE program for expertise and public content sharing in JAX+Flax since August 2022

Tutorials

- Created and taught a series of implementation tutorials in PyTorch and JAX for various Deep Learning topics ([website](#), ~40k page visits per month, >1.9k [GitHub](#) stars). Part of the [DL courses](#) at UvA and [official tutorials](#) of PyTorch Lightning

Teaching assistant

- Teaching assistant (TA) for the graduate courses Deep Learning, NLP 1, FAIR, IR 1, and Advanced Topics in Computational Semantics at the University of Amsterdam (2019-2023)
- Head TA for ASCI PhD Course on Computer Vision 2022 ([link](#))

Summer schools

- Participated in the Oxford ML Summer School 2021, and the CIFAR DL and Reinforcement Learning Summer School 2021

EDUCATION

PhD, Artificial Intelligence

University of Amsterdam, QUVA lab

📅 Sep 2020 – exp. Sep 2024 📍 Amsterdam, Netherlands

- Supervisors: [dr. Efstratios Gavves](#) and [dr. Taco Cohen](#)
- My research focuses on the intersection of causality and deep learning, in particular causal representation learning ([abstract](#))
- [ELLIS](#) PhD; collaboration with Qualcomm AI Research

Master of Science, Artificial Intelligence

University of Amsterdam

📅 Sep 2018 – Aug 2020 📍 Amsterdam, Netherlands

- Courses on AI, including ML, DL, NLP, IR, CV, and RL
- Thesis on "Categorical Normalizing Flows" ([publication](#)), applied for permutation-invariant graph/molecule generation
- Final GPA: 9.5, cum laude (Dutch grading system)

Bachelor of Engineering, Computer Science

Baden-Wuerttemberg Cooperative State University

📅 Oct 2015 – Sep 2018 📍 Stuttgart, Germany

- Cooperative study program with the specialization in IT Automotive and autonomous driving, in cooperation with Daimler AG/Mercedes-Benz R&D
- Final GPA: 1.0 (German grading system)

Pre-Studies, Mathematics/Computer Science

Ruhr University Bochum

📅 Apr 2012 – Sep 2014 📍 Bochum, Germany

- Completing university courses as high school student, including Introduction to Software Engineering I & II, Linear Optimization, and Programming for Mathematicians

AWARDS

- Won 4th place in the NeurIPS 2020 challenge "[Hateful Memes](#)" of Facebook AI ([paper](#))
- Best CS undergraduate student at the Baden-Wuerttemberg Cooperative State University 2018
- Awarded SchülerUni Scholarship 2012/13 for taking university courses on Computer Science/Mathematics during high school

SKILLS

DL Frameworks: PyTorch (2018-present, R&T), JAX/Flax (2021-present, R&T), Tensorflow (2017-2018, R), Caffe (2016-2017, R)
(R - Research, T - Teaching)

Programming languages: Python (2016-present), HTML/PHP/SQL (2012-present), Java (2012-2019), MATLAB (2016-2019), C (2015-2017)

Additional skills: SLURM cluster computing (2018-present), git (2015-present), Docker (2017-present)

SELECTED PUBLICATIONS

Conferences

- Samuele Papa, Riccardo Valperga, David M Knigge, Miltiadis Kofinas, [Phillip Lippe](#), Jan-jakob Sonke, Efstratios Gavves: **How to Train Neural Field Representations: A Comprehensive Study and Benchmark**. The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024 ([link](#))
– *Own contributions*: Core contributor of library development in JAX, proposed and optimized vmap parallelization, advised in model development.
- Davide Talon, [Phillip Lippe](#), Stuart James, Alessio Del Bue, Sara Magliacane: **Towards the Reusability and Compositionality of Causal Representations**. Third Conference on Causal Learning and Reasoning (CLear), 2024 ([link](#)) [[Oral](#)]
– *Own contributions*: Worked on model development, experiment design, and theoretical results.. Contributed to writing.
- [Phillip Lippe](#), Bastiaan S. Veeling, Paris Perdikaris, Richard E. Turner, Johannes Brandstetter: **PDE-Refiner: Achieving Accurate Long Roll-outs with Neural PDE Solvers**. Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS), 2023 ([link](#)). [[Spotlight](#)]
- Sindy Löwe, [Phillip Lippe](#), Francesco Locatello, Max Welling: **Rotating Features for Object Discovery**. Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS), 2023 ([link](#)). [[Oral](#)]
– *Own contributions*: Advised in model development, experiment design and model optimization. Contributed to writing.
- [Phillip Lippe](#), Sara Magliacane, Sindy Löwe, Yuki M. Asano, Taco Cohen, Efstratios Gavves: **BISCUIT: Causal Representation Learning from Binary Interactions**. The 39th Conference on Uncertainty in Artificial Intelligence (UAI), 2023 ([link](#)). [[Spotlight](#)]
- [Phillip Lippe](#), Sara Magliacane, Sindy Löwe, Yuki M. Asano, Taco Cohen, Efstratios Gavves: **Causal Representation Learning for Instantaneous and Temporal Effects in Interactive Systems**. International Conference on Learning Representations (ICLR), 2023 ([link](#)).
- Adeel Pervez, [Phillip Lippe](#), Efstratios Gavves: **Differentiable Mathematical Programming for Object-Centric Representation Learning**. International Conference on Learning Representations (ICLR), 2023 ([link](#)).
– *Own contributions*: Proposed application to object-centric learning. Advised in model development and experimental setups. Contributed to writing.
- Adeel Pervez, [Phillip Lippe](#), Efstratios Gavves: **Scalable Subset Sampling with Neural Conditional Poisson Networks**. International Conference on Learning Representations (ICLR), 2023 ([link](#)).
– *Own contributions*: Advised in model development and experimental setups. Contributed to writing.
- Johann Brehmer, Pim de Haan, [Phillip Lippe](#), Taco Cohen: **Weakly supervised causal representation learning**. Thirty-sixth Conference on Neural Information Processing Systems (NeurIPS), 2022 ([link](#)).
– *Own contributions*: Advised in theory and model development, created datasets, implemented causal discovery pipeline, and contributed to writing.
- [Phillip Lippe](#), Sara Magliacane, Sindy Löwe, Yuki M. Asano, Taco Cohen, Efstratios Gavves: **CITRIS: Causal Identifiability from Temporal Intervened Sequences**. International Conference on Machine Learning (ICML), 2022 ([link](#)). [[Spotlight](#)]
- Anna Langedijk, Verna Dankers, [Phillip Lippe](#), Sander Bos, Bryan Cardenas Guevara, Helen Yannakoudakis, and Ekaterina Shutova: **Meta-learning for fast cross-lingual adaptation in dependency parsing**. Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL, Volume 1: Long Papers), 2022 ([link](#)). [[Oral](#)]
– *Own contributions*: Advised in training and finetuning of large language Transformers, guided as teaching assistant through the project.
- [Phillip Lippe](#), Taco Cohen and Efstratios Gavves: **Efficient Neural Causal Discovery without Acyclicity Constraints**. International Conference on Learning Representations (ICLR), 2022 ([link](#)).
- [Phillip Lippe](#) and Efstratios Gavves: **Categorical Normalizing Flows via Continuous Transformations**. International Conference on Learning Representations (ICLR), 2021 ([link](#)).

Journals

- Sindy Löwe, [Phillip Lippe](#), Maja Rudolph, Max Welling: **Complex-Valued Autoencoders for Object Discovery**. Transactions on Machine Learning Research, Nov 2022 ([link](#)).
– *Own contributions*: Advised in model development, experiment design and model optimization. Contributed to writing
- Douwe Kiela, Hamed Firooz, Aravind Mohan, Vedanuj Goswami, Amanpreet Singh, Casey A. Fitzpatrick, Peter Bull, Greg Lipstein, Tony Nelli, Ron Zhu, Niklas Muennighoff, Rize Velioğlu, Jewgeni Rose, [Phillip Lippe](#), Nithin Holla, Shantanu Chandra, Santhosh Rajamanickam, Georgios Antoniou, Ekaterina Shutova, Helen Yannakoudakis, Vlad Sandulescu, Umut Ozertem, Patrick Pantel, Lucia Specia, Devi Parikh (2021). **The Hateful Memes Challenge: Competition Report**. Proceedings of the NeurIPS 2020 Competition and Demonstration Track, PMLR 133:344-360, 2021 ([link](#)).
– *Own contributions*: Added summary of own solution and results to the Hateful Memes Challenge.

SELECTED PUBLIC TALKS

- 02/2024 Invited Talk at the [CARE Talk Series by Valencelabs](#) on "BISCUIT: Causal Representation Learning from Binary Interactions"
- 02/2024 Invited Talk at the [Bellairs Workshop on Causality](#) on "On Practical Challenges of Scaling Causal Representation Learning"
- 09/2023 Invited Talk at the [AI4Science Talk Series](#) on "Achieving Accurate Long Rollouts with Neural PDE Solvers"
- 02/2023 Invited Talk at the [Rising Stars in AI Symposium](#) at KAUST on "Causal Representation Learning"
- 12/2022 Invited Talk at the [Google Student Developer Club, University of Augsburg](#) on "Machine Learning with JAX and Flax"
- 08/2022 Invited Talk at the [First Workshop on Causal Representation Learning](#) at UAI 2022 on "Learning Causal Variables from Temporal Sequences with Interventions"
- 07/2022 Spotlight Talk at ICML 2022 on our paper [CITRIS: Causal Identifiability from Temporal Intervened Sequences](#)
- 07/2022 Contributed Talk at the [Workshop On eCommerce](#) at SIGIR 2022
- 07/2021 Contributed Talk at the [8th Causal Inference Workshop](#) at UAI 2021
- 12/2020 Contributed Talk at the [NeurIPS 2020 Hateful Memes Competition](#) workshop on our 4th place winning solution

REVIEWING

I have served as reviewer for the following conferences and workshops: [ECCV-2020](#), [ICCV-2021](#), [CausalUAI-2021](#), [ICLR-2022](#), [CVPR-2022](#), [CLear-2022](#), [NeurIPS-2022](#), [CRL-2022](#), [CML4Impact-2022](#), [CDS-2022](#), [nCSI-2022](#), [ICLR-2023](#), [CLear-2023](#), [TSRL4H-2023](#), [Physics4ML-2023](#), [UAI-2023](#), [Frontiers4LCD-2023](#), [NeurIPS-2023](#), [ICML-2024](#).